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Introduction



It is important to test computer networks to assure that these networks will be effective especially in the long run.

Aside from ensuring that these networks are secure, it is essential that we know in the first place that they are working correctly to avoid issues in the future.

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TOPIC 1: Network Testing



Network Testing is the actual measurement of the network's state of operation over a period of time. This record will be the basis for comparison or control.

It is often called **understanding the past to predict the future**.

A good **network testing strategy** is one where the network manager is able to predict the operation of the network.

Testing the reliability and its ability to support new applications and users are among the list of factors in examining a good network.

TOPIC 2: Important Network Characteristics

Through a comprehensive network test, the network manager is able to **maintain** a network actively. When it is implemented properly, it is ***easy to track the changes and trends*** that can be seen in its daily operation.

From here, a network manager is able to obtain **valuable insight** where s/he may use for further improvements and troubleshooting.

We should look for the following when we are testing the network:

- (1) Utilization levels
- (2) Number of users
- (3) Number of operational protocols
- (4) Error statistics
- (5) Application utilization

TOPIC 3: Final Testing



Final testing aims to test the hardware and software aspect of the systems and the network.

Follow the following steps.

- (1) Launch **cmd** by CTRL+R, typing cmd, then press enter.
- (2) Type **ping** and the IP address of a computer in the network. If the timeout does not request, it means that the computer is connected to the network.
- (3) You can log-on to the network as well by accessing your network device. Observe the color of the NIC on the computer; the light should be green or yellow and has one solid and a blinking light.

TOPIC 4: Generating Reports



Network testing will essentially need documentation and **reports** from the computer. This is important since this will be used for maintenance.

The following steps may require you to be an **Administrator** to access some of the features in this procedure.

To enable **diagnostic logging**,

- (1) Open the Network Connections folder and view the available connections.
- (2) Press ALT to display the menu bar, then access *Advanced*, click *Remote Access Preferences*.
- (3) Access *Diagnostics*, you can do one or more of the following:
 - a) You can enable logging for all remote access connections. Select *Enable logging*. Then click *Generate Report*.
 - b) You can clear logs that were created during previous connection attempts. Select *Clear Log*. Then click *Generate Report*.

To view this report, click *View Report*.

To save it to a file, click *Save report to this file*.

To send report to an e-mail address, click *Send report to this e-mail address*.

To generate a more concise report, click *Generate summary report*.